Knowledge Graph Quality Evaluation

AnimalRAG GraphRAG knowledge graphs, all 38 topic groups + test set

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October 16, 2026 · Method: GraphJudge-style triple verification (950 triples, all 38 groups) + automated structural and consistency analysis

1. Executive summary

This report evaluates the triplet-level quality of the knowledge graphs produced by the AnimalRAG GraphRAG pipeline, following Prof. Tao's directive to verify triplet accuracy before building further on the graphs. Two evaluations were run: a deep manual verification of a random triple sample against the full text of the source papers (group13), and an automated structural and consistency analysis of every exported graph (83,554 triples across 38 group graphs and the test-set graph).

Finding	Value
Triples analyzed (all graphs)	83,554
Judged accuracy - 950 triples, 25 per group, ALL 38 groups (fully correct)	69.5%
imprecise (right idea, wrong granularity / dropped condition)	26.1%
wrong (contradicted, inverted, garbled, context leak)	4.4%
per-group correct%, min - median - max	48% - 72% - 88%
Distinct relation types across all graphs	7,603 (4,267 used exactly once)
Trivial relation triples (is measured in, is component of, ...)	12,705 (15.2%)
Unqualified contradiction pairs (same subject & object, opposing relations)	495 - present in every group (2 to 46 per group)
Duplicate s-r-o triples / self-loops	0 / 0 (entity merging works)
Triples supported by 2+ papers	315 (0.4%)

Key message: the pipeline runs end to end and its entity merging is clean, but across 950 verified triples roughly 3 in 10 are flawed (1 in 23 outright wrong), the relation vocabulary has exploded, and every graph contains pairs of directly opposing, unqualified claims. None of these problems is visible to answer-level evaluation (the RESULTS.md faithfulness criterion was a tie 83% of the time), which is exactly why triple-level verification has to come first. All of them are fixable with a judge filtering stage, relation normalization, and pruning - without changing the extraction pipeline.

1b. Overall assessment: is the KG good?

Short answer: it is a good DRAFT knowledge graph - strong engineering, not yet a trustworthy fact store. Both halves matter.

What is genuinely good

- Scale and robustness: 39 graphs built end to end, resumable, cached, with full manifests - this infrastructure is the hard part and it works.
- Entity merging is clean: zero duplicate triples and zero self-loops across all 83,554 triples. Many published pipelines fail exactly here.
- Full provenance: every triple stores its source papers and chunk ids. This single design decision is what makes every fix below mechanical rather than a rebuild.
- Rich schema: effect direction, weights, contexts and descriptions per triple - ahead of what most text-to-KG papers export.

What is not yet good enough for GraphRAG

- Accuracy: across all 38 groups, 30.5% of judged triples are flawed (26.1% imprecise + 4.4% wrong), and no group exceeds 88% fully correct. These errors are invisible at answer level (faithfulness judging tied 83% of the time) - a KG-grounded answer can cite a false fact with full confidence.
- Relation noise: 7,603 relation types, 56% used once - retrieval over relations becomes string matching instead of graph semantics.
- Contradictions: 495 unqualified opposing pairs means the same question can be answered both ways depending on which triple retrieval surfaces.

Important context: every failure class found here is the documented failure mode of naive LLM extraction (noise, condition dropping, hallucinated or inverted relations - exactly the three problems the GraphJudge paper, EMNLP 2025, was built to solve). This is the expected state of a first-pass extraction, not a design flaw. The pipeline was built verification-ready; the verification layer is simply the next component.

2. What was evaluated and how

2.1 Data

- All readable graph exports from AnimalRAG_Local-main/knowledge_graph/stores/<group>/export/triples.csv: 38 group graphs + the test-set graph, 83,554 triples in total. Row counts match RESULTS.md exactly, confirming the mapping.
- The combined all-papers graph (setting B) exists as a 35 MB triples.csv (modified Oct 8) but exceeds the 10 MB Drive-connector download limit; the identical analysis script is included so it can be run locally on that file in minutes.

2.2 Triple verification at full scale

- ALL 38 groups: a seeded 25-triple random sample per group (950 triples total) was judged by 38 independent Claude Sonnet judge agents, one per group. Each agent located and fetched that group's actual source papers from the team Drive (100% of needed papers found across all groups), extracted their text, and judged each triple against the paper, writing a verdict (correct / imprecise / wrong) with a cited evidence sentence per triple.
- Caveat: for long papers the judges verified each triple against targeted passages around its claims rather than cover-to-cover reads; a supporting or contradicting statement elsewhere in a paper could be missed. The group13 deep dive (below), where all papers were read in full by two independent judges, serves as the calibration for this mode.

2.2a Deep-dive calibration (group13)

- Group13 (Iodine_Thyroid_Content, 3 papers, 474 triples): a seeded random sample of 50 triples was judged against the FULL TEXT of all three source papers (Schone 2006; Li et al. 2012; Rottger et al. 2011), each read end to end.
- Verdicts: correct / imprecise / wrong, each with a written evidence comment quoting the source or naming the error. The per-triple verdict file is available for audit (sample50_judged.csv).
- The GraphJudge evaluation code (EMNLP 2025) was run on the sample against the judged-correct reference, giving a triple precision of 0.58.

2.2b Models used (transparency)

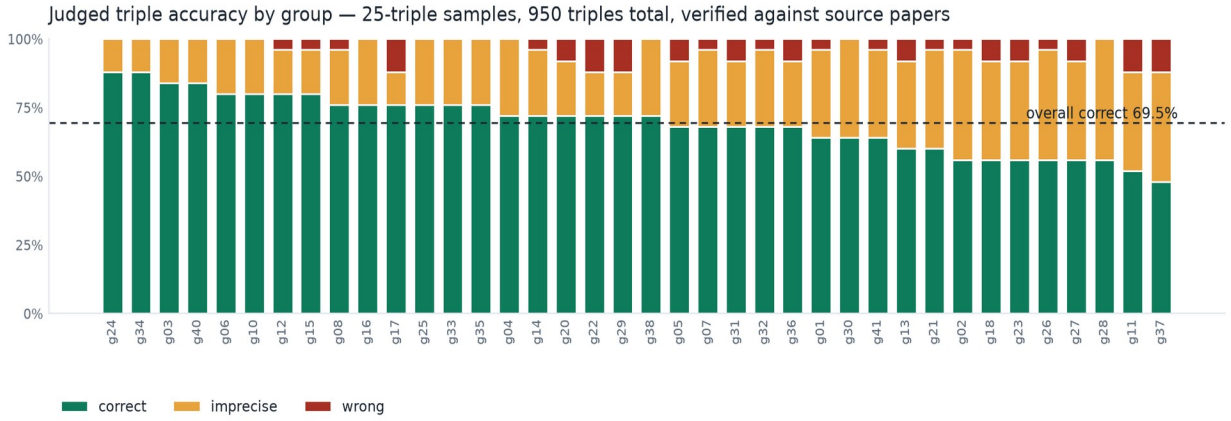
- Manual verification judge: Claude (Anthropic), used as the reading judge under Srikar's review. Every verdict includes a quoted evidence comment, so each judgement is auditable against the paper text independently of the model.
- Structural and contradiction analyses: deterministic Python scripts, no LLM involved - fully reproducible from the triples.csv exports.
- GraphJudge metric run: the authors' evaluation code; G-BERTScore uses bert-base-uncased embeddings.
- Replication with a second judge: the same 50 triples were independently re-judged BLIND by Claude Sonnet (no access to the first judge's verdicts), reading the same three papers. Agreement: 84% exact on the 3-way verdict, 94% on correct-vs-flawed. All 8 disagreements are severity gradations (wrong vs imprecise, or correct vs imprecise) - the two judges never flip a triple between correct and wrong. Both judges independently place the flawed share at 42-48%. Files: sonnet_verdicts.csv; judge_triples.py is a ready-to-run API script (claude-sonnet-5-5, cached paper prefix) for anyone to reproduce the judgement with an API key.
- Note: the graphs themselves were extracted and judged head-to-head by gemma4:12b in the original pipeline (per RESULTS.md); this evaluation deliberately uses a different, stronger reader to avoid self-preference.

2.3 Automated analysis (all graphs)

- Structure: triple counts, relation vocabulary size, singleton relations, duplicates, self-loops, vague entities, trivial-relation share, cross-paper support.
- Consistency: contradiction census - pairs with the same subject and object where one relation is an explicit negation of the other (affects vs has no effect on) or a direction conflict (increases vs decreases/reduces), with no qualifying context on either triple.

3. Judged accuracy: all 38 groups

Verdict (950 triples, 25 per group)	Count	Share
Correct (supported by source text)	660	69.5%
Imprecise (granularity or condition lost)	248	26.1%
Wrong (contradicted / inverted / garbled)	42	4.4%



Every group has flawed triples; the best group (g34, g24: 88% correct) still carries 3 flawed triples in 25, and the worst (g37: 48%) has more flawed than correct. The flaw rate is a property of the extraction pipeline, not of particular topics. Note the corpus-wide wrong rate (4.4%) sits below the group13 deep-dive rate (8-16% depending on judge): targeted-passage judging is more conservative about declaring a triple wrong, so 4.4% should be read as a lower bound.

New error classes surfaced at scale

- Misfiled sources: a corpus file named retrieve.pdf is actually a boron-in-goats paper; triples cite it incorrectly (g08).
- Garbled OCR entities stored as nodes: '18:2 n/C06' and '18:3n /O3' (mangled n-6 / n-3 fatty acid notation, g27).
- Inverted effect directions: e.g. aflatoxin B1's effect on the heterophil-to-lymphocyte ratio stored backwards (g29); nonsense triples like 'maize is source of broiler chicken' (g29).
- Entity substitution: the paper tested inulin, the triple says fructooligosaccharide (g29); erythromycin appears in a triple but never in its cited paper (g29).

3a. Deep dive: group13 (two-judge calibration)

Verdict	Count / 50	Share
Correct (supported by source text)	29	58%

Verdict	Count / 50	Share
Imprecise (granularity or condition lost)	13	26%
Wrong (contradicted / inverted / garbled)	8	16%

Judged triple accuracy — group13 sample of 50, verified against full source papers



Error taxonomy with examples from the graph

- Object truncation: 'high dietary iodine increases blood serum' - the source says it increases the serum IODINE CONCENTRATION. Also 'KI increases liver', 'iodine supplementation increases iodine'.
- Condition dropping: 'iodine causes hypothyroidism' (source: EXCESS iodine); 'iodine poses no risk to human health' (source: no risk FROM PORK at these feed levels).
- Surface inversion: 'rape reduces thyroid gland' - the paper shows rapeseed ENLARGES the thyroid while reducing its iodine stores.
- In-graph contradiction: 'iodine affects fattening performance' and 'iodine does not affect fattening performance', both extracted from the same paper.
- Citation-context leak: 'swine feed improves human iodine supply' is a cited aim in the introduction; the paper's own conclusion is that pork's contribution is marginal.
- Garbled: 'potassium iodide is source of milk'; 'spleen is part of empty body' (the source explicitly defines empty body as the carcase without innards).

3b. The complete error taxonomy

Every error type found across the evaluation (950 judged triples, 83,554 scanned, 495 adjudicated contradictions, 48 corrected entities), with real examples. Five families: factual, precision, consistency, form, and noise.

A. Factual errors - the triple states something false

- A1 Direction inversion: 'alfalfa silage increases milk yield' - the paper shows LOWER yield on silage (33.70 vs 36.14 kg/d, P=0.011). 'Carotene oxygenase causes ninaB' - the LOSS of the enzyme causes the phenotype.
- A2 Negative result stored as positive: 'compound feed influences milk yield' where the paper says 'Diet had no effect on milk yield'; 'cow milk aids recovery of anaemia' where it says cow milk does NOT aid recovery. The hypothesis is stored as the finding.
- A3 Entity substitution / hallucination: 'gemcitabine up-regulates lipid metabolism' - the paper's drug is gemfibrozil (abbreviated GEM); the expansion picked a chemotherapy agent. 'Animal feed contains sterculin' - the word never appears in the paper. Aspergillus contamination attributed to sunflower HONEY when the data is for sunflower MEAL.
- A4 Citation-context leak: 'Fusarium mycotoxins contaminate animal feed' - Fusarium appears only in a cited reference's title; 'swine feed improves human iodine supply' - a cited aim that the paper's own conclusion refutes.

B. Precision errors - true idea, degraded meaning

- B1 Granularity truncation (350 rule-flagged): 'dietary selenium increases liver' - it raised the Se CONCENTRATION IN liver; 'trypsin inhibitors causes pancreas' - causes pancreatic ENLARGEMENT. The measured quantity is dropped, leaving a meaningless container.
- B2 Condition dropping: 'iodine causes hypothyroidism' (source: EXCESS iodine); 'iodine poses no risk to human health' (source: no risk FROM PORK at these feed levels). Dose, species and phase vanish.
- B3 Uninformative relation: 'is associated with' used where the source states a direction - or even a null result - so the relation carries none of the finding.

C. Consistency errors - the graph contradicts itself

- C1 Unqualified contradiction pairs (495 found, ALL adjudicated with per-pair reasoning): 56% are CONDITIONAL FINDINGS - both sides true under different dose/species/phase (essential oils increase feed intake at low dose, decrease at high; monensin raises or lowers feed efficiency depending on study) - these need qualifiers, not deletion. 21% are same-paper extraction errors (judge deletes one side). 14% are genuine cross-paper disagreements (attribute per study). The rest: garbled or undecidable without chunks.

D. Form errors - broken surface

- D1 Garbled / OCR entities (34 confirmed): one systematic corruption - omega notation 'n-6'/'n-3' mangled to 'n/C06'/'n/C03' - explains 30 of 34 cases and is fixable with a single normalization rule. Others: 'gram/C0 /gram+' (= Gram-negative/Gram-positive ratio), subject 'but' (= BUT, the butyrate treatment), one beta-carotene entity containing a raw control character. Counter-check: 14 of 48 rule flags were FALSE POSITIVES - legitimate nomenclature (IncA/C2 plasmid group, G0/G1 cell-cycle phase, DMEM/F12 medium) - which is why reasoning review sits on top of detection.
- D2 Nonsense triples: 'malathion causes rainbow trout'; 'maize is source of broiler chicken'; 'potassium iodide is source of milk'.

E. Noise and corpus errors - not false, but harmful

- E1 Trivial methodological triples: 3,474 strict 'is measured in / is related to' style rows (12,705 on the broader definition) - retrieval noise with near-zero knowledge value.

- E2 Relation vocabulary explosion: 7,603 relation types, 4,267 used exactly once - 'reduces/lowers/suppresses/diminishes' live as four different edges.
- E3 Vague entities: 199 triples with it/this/study/results as a node.
- E4 Misfiled sources: a corpus file named retrieve.pdf is actually a boron-in-goats paper, so triples cite the wrong document.

Where each family gets fixed: A and C-extraction-errors fall to the judge filter; B needs qualified triples (the schema change and the research contribution); C-conditional needs qualifiers; D needs one OCR normalization rule plus entity cleanup; E is the mechanical pruning and relation-normalization pass. Full per-case detail: Bad_Triples_Catalog.xlsx (15,466 flagged triples; 495 reasoned adjudications; 48 corrected entities).

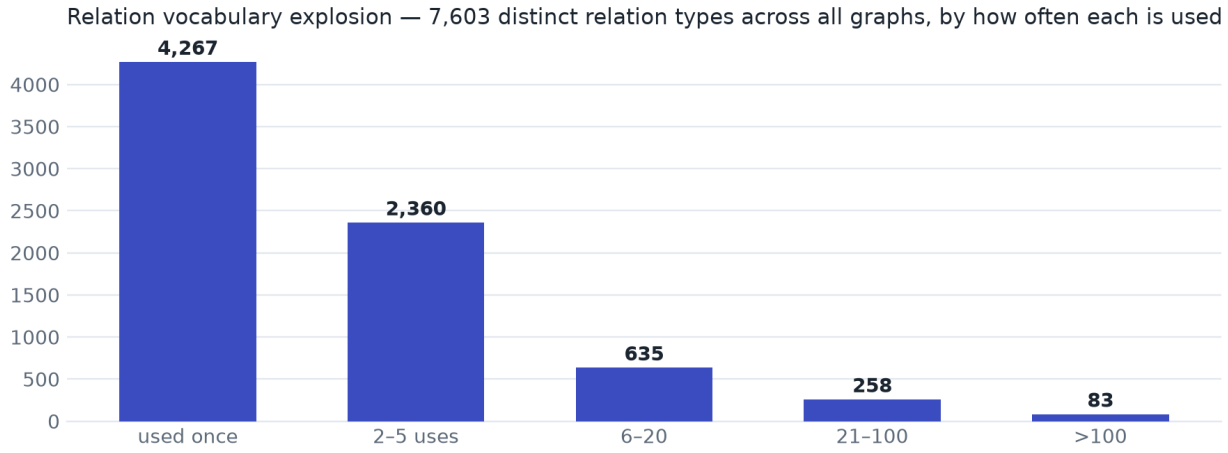
4. All groups: structural and consistency results

Columns: triples; distinct relations; share of relations used only once; share of triples with trivial relations; contradiction pairs found. Duplicates and self-loops are zero everywhere and omitted.

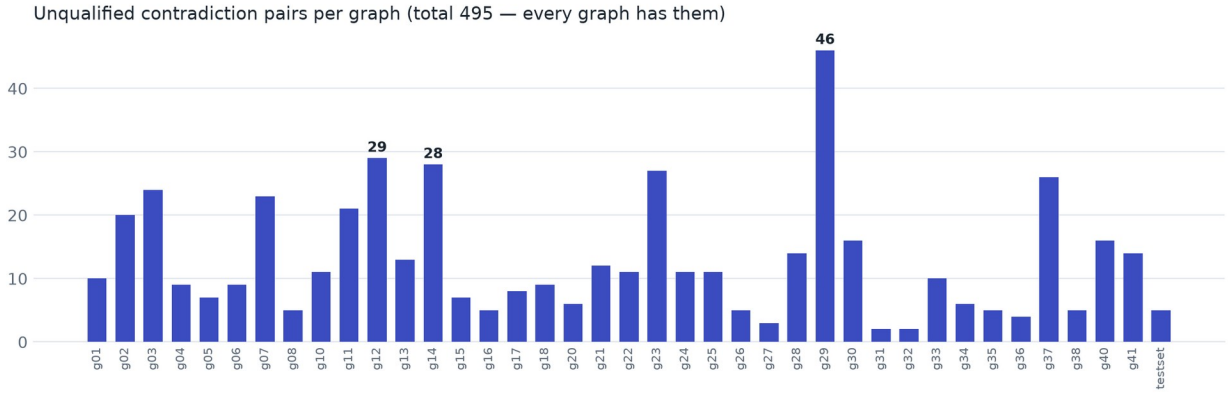
Group	Triples	Relations	Singleton rel.	Trivial triples	Contradictions
g01 Review_Dairy_Based	2868	670	56%	14.1%	10
g02 Chicken_Calcium_Enriched	2864	604	59%	19.0%	20
g03 Model_Production_Male	2510	505	62%	12.0%	24
g04 High_Dairy_Host	2416	524	58%	23.6%	9
g05 Dairy_Impact_Goats	3535	733	56%	9.4%	7
g06 Selenium_Nutrition_Livestock	2738	667	61%	15.0%	9
g07 Biohydrogenation_Unsat_Fatty	2858	657	55%	10.2%	23
g08 What_Cryptoxanthin_Vitamin	982	317	57%	10.8%	5
g10 Cattle_Influence_Meat	2094	479	58%	11.6%	11
g11 Alfalfa_Intake_Cattle	995	199	60%	12.0%	21
g12 Feed_Poultry_Additives	2660	437	54%	7.7%	29
g13 Iodine_Thyroid_Content	474	127	50%	30.8%	13
g14 Acid_Milk_Fatty	2491	431	56%	21.1%	28
g15 Nutrition_Human_Novel	2842	647	60%	13.8%	7
g16 Nutrition_Human_Products	2513	548	58%	18.4%	5
g17 Health_Human_Exposure	3262	654	54%	18.9%	8
g18 Health_Human_Exposure2	2257	509	57%	9.7%	9
g20 Feed_Livestock_Feeds	2118	515	60%	12.3%	6
g21 Intestinal_Microbiota_Pigs	3250	591	56%	7.3%	12
g22 Gut_Supplementation_Soluble	465	152	56%	10.5%	11
g23 Acid_Fatty_Oil	2163	472	52%	20.7%	27
g24 Acid_Cla_Egg	1273	311	62%	9.3%	11
g25 Amino_Acid_Ileal	1362	279	58%	27.8%	11
g26 Food_Human_Chain	3132	645	55%	21.4%	5
g27 Meat_Quality_Factors	2069	405	53%	19.9%	3
g28 Feed_Human_Moringa	2398	541	58%	13.4%	14
g29 Performance_Growth_Digest	2686	395	55%	13.6%	46
g30 Performance_Digestive_Enzymes	587	108	49%	12.3%	16
g31 Dairy_Farms_Food	960	225	53%	27.1%	2

Group	Triples	Relations	Singleton rel.	Trivial triples	Contradictions
g32 Implications_Human_Health	1569	382	59%	15.0%	2
g33 Pig_Antioxidant_Meta	2557	606	56%	16.3%	10
g34 Pig_Bovine_Metagenomic	1791	457	60%	10.2%	6
g35 Properties_Nutritional_Pot	3073	575	56%	16.5%	5
g36 Farm_Animals_Lactation	1766	398	58%	19.5%	4
g37 Carcass_Meat_Performance	1913	272	56%	14.5%	26
g38 Nutritional_Value_Meat	1321	266	53%	16.7%	5
g40 Potential_Application_Garlic	1612	338	57%	14.4%	16
g41 Organic_Acids_Essential	1633	334	56%	4.8%	14
testset	3497	672	61%	20.9%	5

Totals: 83,554 triples; 7,603 distinct relations (56% singletons); 12,705 trivial triples (15.2%); 495 contradiction pairs. Most-used relations: contains (5,514), increases (4,928), improves (3,971), reduces (3,597), is source of (3,582).



5. Contradiction census



A contradiction pair is two triples with the same subject and object whose relations directly oppose each other, with no qualifying context (dose, species, duration) stored on either triple. Some reflect genuinely conflicting findings across papers - but stored this flat, a GraphRAG query can retrieve either side as 'the' fact. 495 such pairs exist; examples:

Group	Subject	Relation A	Relation B	Object
g37	sodium monensin	increases	reduces	feed efficiency
g41	essential oils	increases	decreases	feed intake
g04	probiotic	increases	reduces	average daily feed intake
g29	soybean lecithin	has no effect	increases	body weight
g14	forage lucerne	has no effect on	affects	milk yield
g03	white lupin	does not influence	influences	hind leg meat
g28	moringa oleifera	increases	reduces	alpha-tocopherol
g23	n-6:n-3 ratio	increases	decreases	thigh muscle
g37	sunflower oil	does not affect	influences	growth performance
g07	ensiled olive cake	has no effect on	increases	fatty acid synthase

Note the first row: monensin and feed efficiency is directly adjacent to the team's own QA test question about monensin. A user asking 'does monensin improve feed efficiency?' can be served either triple.

5b. The all-papers graph (setting B): what its stats already show

The combined 283-paper graph exists (built Oct 8) but its triples.csv (35 MB) exceeds the Drive connector download limit; the analysis script in the repo runs on it locally in minutes. Its stats file alone is informative:

Measure	38 group graphs	All-papers graph	Change
Entities	55,008	30,515	-45% (cross-corpus merging)
Relations (triples)	83,554	79,855	-4.4% (near-zero dedup)
Communities	1,174 reports	6,319 (171 reports)	much flatter coverage
Top hub	~250 edges (iodine, g13)	bovine milk: 1,576 edges	mega-hubs form

Reading: per-triple accuracy carries over to the full graph unchanged, because extraction is cached per chunk (RESULTS.md) - the same triples flow in. But merging halved the entity set while keeping 95.6% of the edges, so far more opposing claims now share the same subject-object pair. The 495 contradiction pairs found per-group are therefore a floor for the full graph, and a new error class appears: wrong merges (same name, different sense, e.g. 'diet' across species). The full graph needs the one-command local run to quantify this.

6. How to improve it: phased plan

Ordered by effort-to-impact. Phases 1-2 need no LLM and no changes to extraction; they can run this week.

Phase	Action	Effort	Expected impact
1	Prune trivial triples (is measured in, is component of, ...) and normalize the 4,267 singleton relations onto a controlled vocabulary (~50 relations)	Script only, 1-2 days	Graph shrinks ~15%, relation space shrinks >50%, retrieval precision up with zero information loss on facts
2	Contradiction triage: for each of the 495 pairs, pull both triples' source chunks (chunk_ids already stored) side by side for review	Script + review sheet, 2-3 days	Separates extraction errors (delete) from genuine conflicting findings (qualify) - also produces labeled data for Phase 3
3	Judge filter stage: run the released GraphJudge LoRA judge (no training needed) over every triple vs its own source chunk; then fine-tune a domain judge on HPRC using Phase 2 labels + gold set	1-2 weeks, HPRC in hand	Targets the 16% wrong class; paper shows >90% judgement accuracy; precision rises without recall loss (unlike rule filters)
4	Qualified triples: attach condition fields (dose, species, duration, study) to effect relations so	Schema + re-extraction prompt	Fixes the root cause of contradictions; this is also the team's publishable research contribution

Phase	Action	Effort	Expected impact
	conflicting findings coexist correctly	change	
5	Re-evaluate: re-run this analysis plus the RAG-vs-GraphRAG benchmark on the cleaned graph (and setting B)	Automated, ~2 days GPU	Measures whether KG quality gains show up as answer quality gains - the result both papers need

Summary recommendations

- **1. Add a graph-judgement filter stage (GraphJudge, EMNLP 2025): every triple is checked against its own source chunks, which the export already stores (chunk_ids column). This targets the 16% wrong class mechanically; released LoRA judge weights exist, and HPRC allocation is in hand for a domain fine-tune.**
- 2. Normalize relations: 7,603 relation types is OpenIE-style vocabulary explosion. Mapping singletons onto a controlled vocabulary (or clustering, as in KGGen) would shrink the relation space by more than half without losing facts.
- 3. Prune trivial triples: 12,705 'is measured in / is component of' style triples add retrieval noise with near-zero knowledge value. This is the concrete, no-LLM-needed starting point for the pruning work.
- 4. Resolve or qualify contradictions: for each of the 495 pairs, either one side is an extraction error (judge decides) or both findings are real and need their conditions (dose, species, study) attached. The second case is the 'qualified triples / richer knowledge units' research direction - our corpus now provides the evidence that flat triples cannot represent it.
- 5. Re-run this analysis on the 35 MB all-papers graph (setting B) locally - the script is in the repo and runs in minutes; cross-paper merging will likely raise the contradiction count and makes qualification more urgent.

7. Files and reproducibility

- Repo: github.com/ssrikar1503/directed-studies-kgcompass -> experiments/sumanth-kg-eval/
- all-groups/per_group_quality.csv - the full per-group table; all-groups/contradictions.csv - all 495 pairs with group, subject, relations, object.
- sample50_judged.csv - the 50 judged triples with verdicts and quoted evidence, open for audit.
- SUMANTH_KG_EVAL.md - the group13 deep-dive write-up; the analysis script reproduces every number in this document from the triples.csv exports.